

LEDGER · 2026-08-29 · MATERIALS SCIENCE / CHEMICAL ENGINEERING

PDMS/SiO2 Superhydrophobic Anti-Soiling Nanocoating for Photovoltaic Arrays

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	ChemiTek Solar Wash Protect (SWP)
Entry application	Utility-Scale and Commercial C&I Solar Operations & Maintenance (O&M) Contractors performing scheduled retrofit cleaning
Measured advantage	1.0x delta (~5% power output increase vs ChemiTek SWP's up to 5% power output increase over standard water washing)
Time to first revenue	3 months
Gross margin at parity	84.4%
Startup CapEx	\$115,000
Payback from first sale	0.4 months

Capital productivity

44.66x

Regulatory risk

Moderate — qualification costs reported (\$25,000 estimated)

Why it failed

- Voided by: Mismatched comparison, Negative control / sub-2x, Form-factor / handling regression
- Below the 2x step-change bar

Assessment

The concept fails the superiority threshold with a 1.0x performance delta (~5% efficiency gain vs ChemiTek SWP's 5%). Additionally, it introduces severe form-factor and hazard regressions by requiring toxic TEOS, flammable ethanol, and ABEK respirators compared to the incumbent's safe, water-based additive.

Also flagged

- unsourced_equipment
- no_opex

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-29 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

Free studies

Track record

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